

Configuring an Active Directory Home Lab

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Overview

We will be creating a lab environment utilizing several Windows 10 VMs along with Windows Server 2022. Windows Server 2022 will be used as our domain controller, which will allow for authentication and authorization for users that we establish within AD. The overall goal is to get acquainted with Active Directory, understanding what it provides for us and how it works. Active Directory is widely used in Windows domain enterprises, you will need to understand it at some point in your IT career.

Lab Specs & Notes

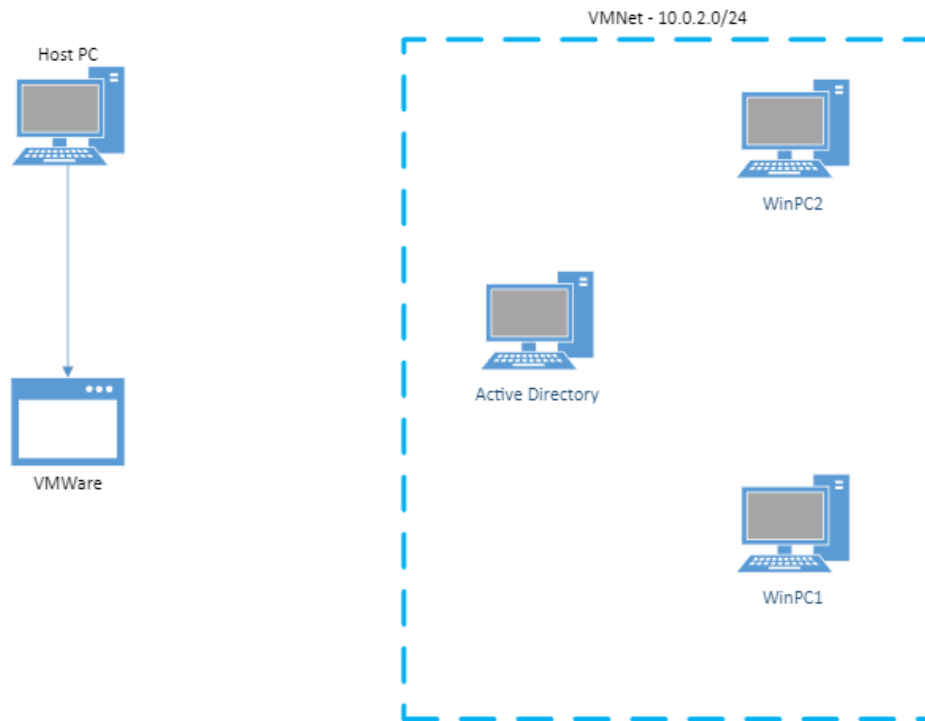
Depending on your specs, and how many VMs you would like to have, you can upscale this lab as much as you want. For my purpose, I will only be utilizing 2 Windows 10 hosts along with Windows Server 2022. This will provide the core elements necessary to mimic a traditional on-premises enterprise network, despite the lack of hosts. You have the ability to add additional systems later on if you choose.

Host PC Specs:

- Windows 11
- AMD Ryzen 7 5800
- 16 GB RAM
- 1 TB SSD

Virtual Machine Specs:

- Active Directory - Windows Server 2022 (1 CPU, 2 GB RAM)
 - <https://www.microsoft.com/en-us/windows-server>
- Workstation 1 - Windows 10 Enterprise (1 CPU, 2 GB RAM)
 - <https://www.microsoft.com/en-us/software-download/windows10>
- Workstation 2 (optional) - Windows 10 Enterprise (1 CPU, 2 GB RAM)



Note: Does not have to be VMWare,
use your preferred hypervisor

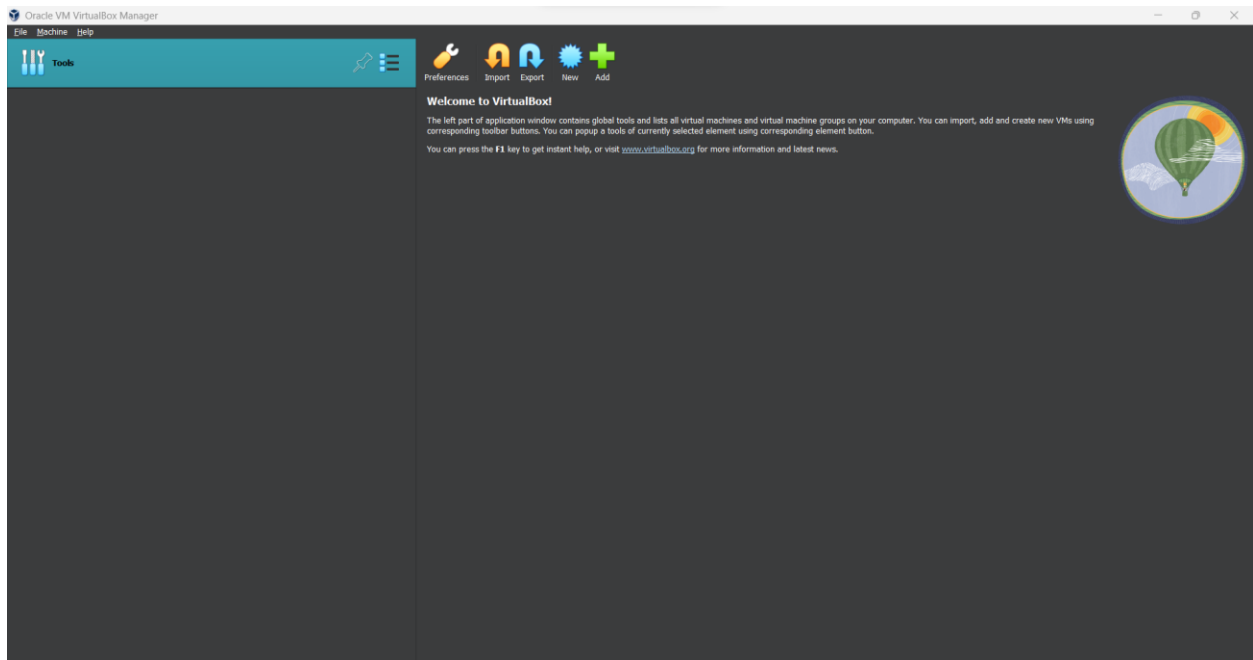
Other Notes

- I am using [VMware Workstation 17 Pro](#) as my hypervisor, but there are plenty of free options that work just as well. I recommend [VirtualBox](#) as it has a lot of the same features as VMware and will work just fine for this lab.
- All the steps below will be roughly the same for every hypervisor with the exception of buttons being in different locations. If you have trouble locating what I am referring to, do a google search. There is lots of documentation for all the popular hypervisors.
- **IMPORTANT:** You need to make sure that you have virtualization enabled in your BIOS. If you do not have it enabled, you will not be able to start the VMs. To enable it, google your laptop or motherboard model depending on if you have a desktop and search "how to enable virtualization in x bios." (x being your motherboard or laptop model)

Configuring Your Hypervisor

1. After you have downloaded and installed your hypervisor, boot it up. I will be using VirtualBox for my demonstration images for configuring the VMs as my VMware installation is already populated with VMs.

You will be greeted with this (or similar):



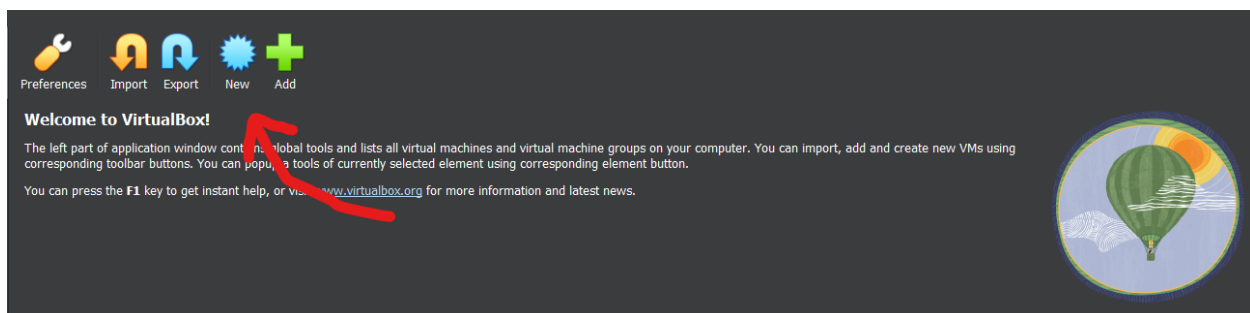
2. We will begin by installing Windows 10 Enterprise. Here is the download link if you haven't already got it: <https://www.microsoft.com/en-us/evalcenter/download-windows-10-enterprise>

Choose your respective language and click the 64-bit edition, the download will start immediately.

Please select your Windows 10 Enterprise download

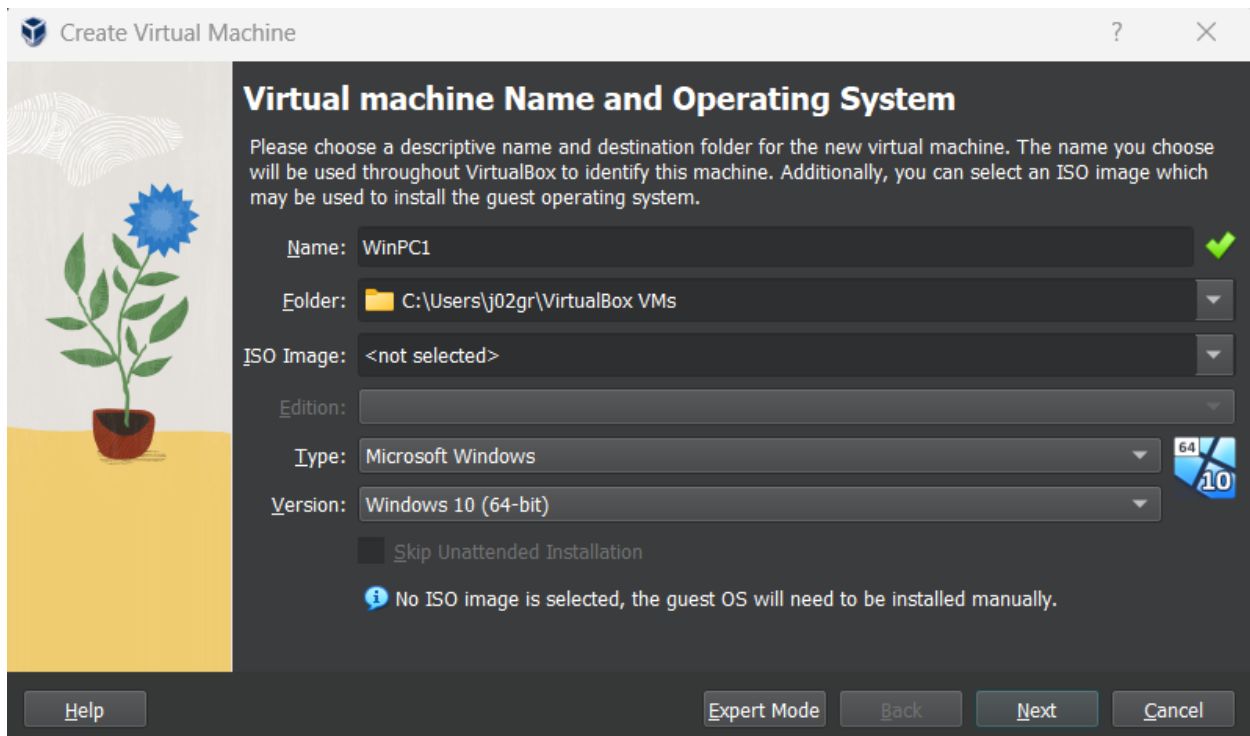
English (United States)	ISO – Enterprise downloads 32-bit edition > 64-bit edition >	ISO – Enterprise LTSC downloads 32-bit edition > 64-bit edition >
English (Great Britain)	ISO – Enterprise downloads 32-bit edition > 64-bit edition >	ISO – Enterprise LTSC downloads 32-bit edition > 64-bit edition >

- Now that you have downloaded the ISO, we can set up the VM. Head over to your respective hypervisor, there will be a button to “Add a Virtual Machine” or similar. For the case of Virtual Box, click “New.”

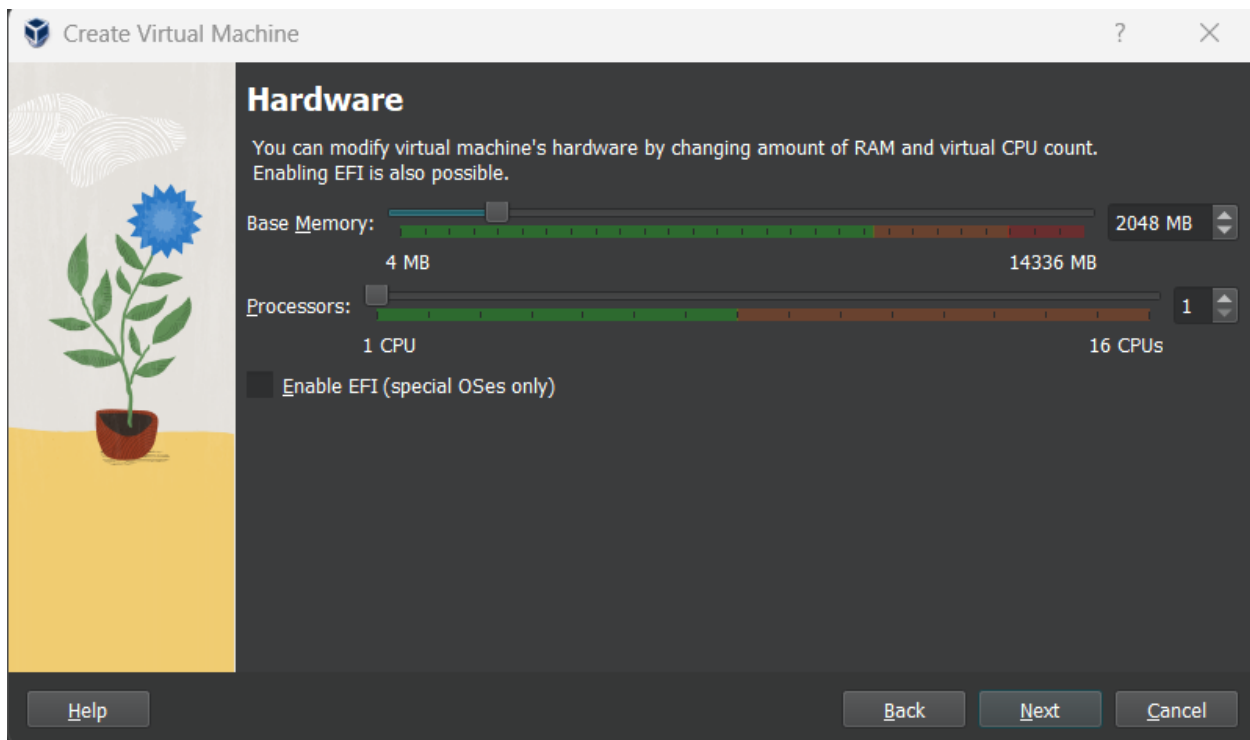


- After clicking “New,” a box will pop up. In the image below, I’ve included all the settings you should have for this first page. Feel free to choose whatever naming scheme you would like for your VMs.

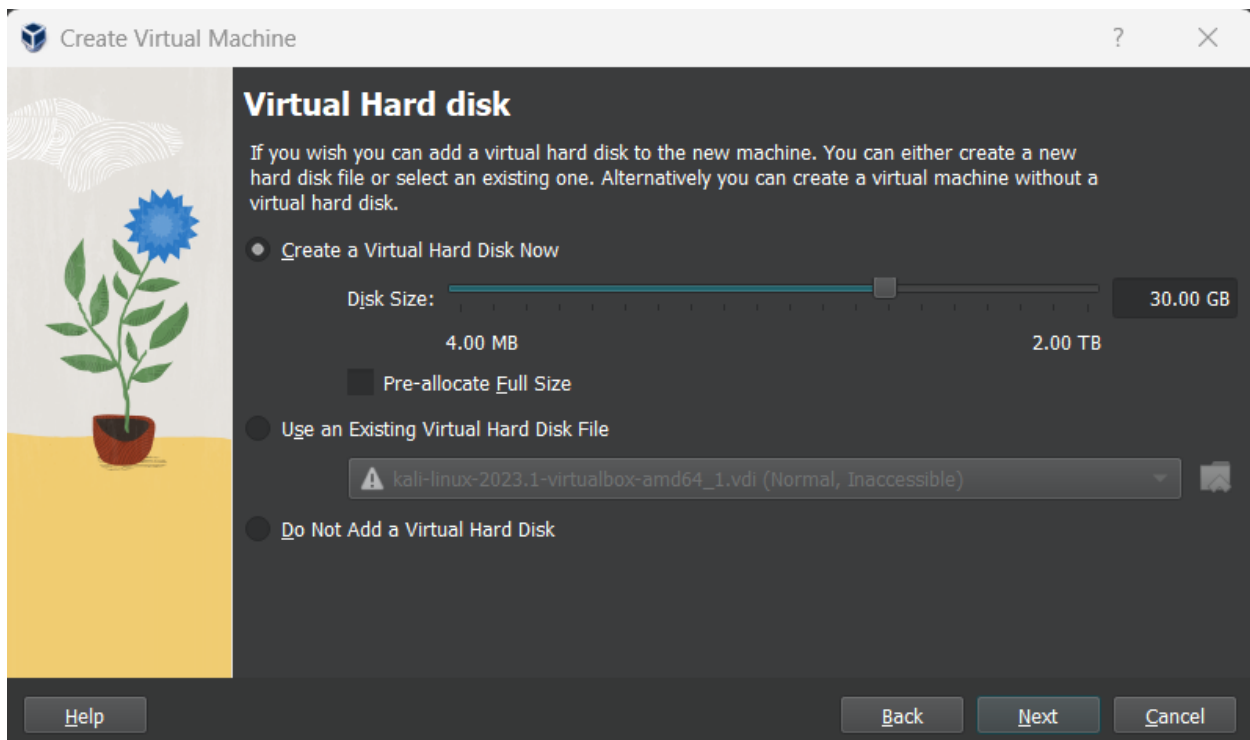
IMPORTANT: Do not add the ISO, we will add it once we have configured the Virtual Machine. If you add it in the initial setup, you will not be able to boot into the installation.



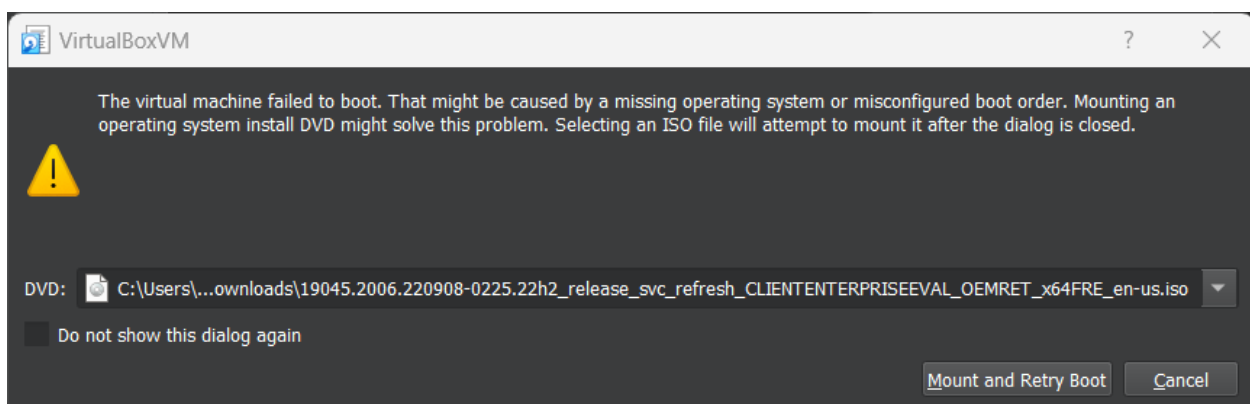
5. After you input the details above, click “Next.” We will be allocating 2 GBs of RAM, and 1 CPU. Depending on your system, you can allocate more RAM or CPUs. I do not recommend going below 2 gigs of RAM.



- Now, we will be creating a virtual hard disk. The minimum disk space for Windows 10 is 16 GBs, but I will be allocating 30. Feel free to add more or less, just do not go below 16 gigs.

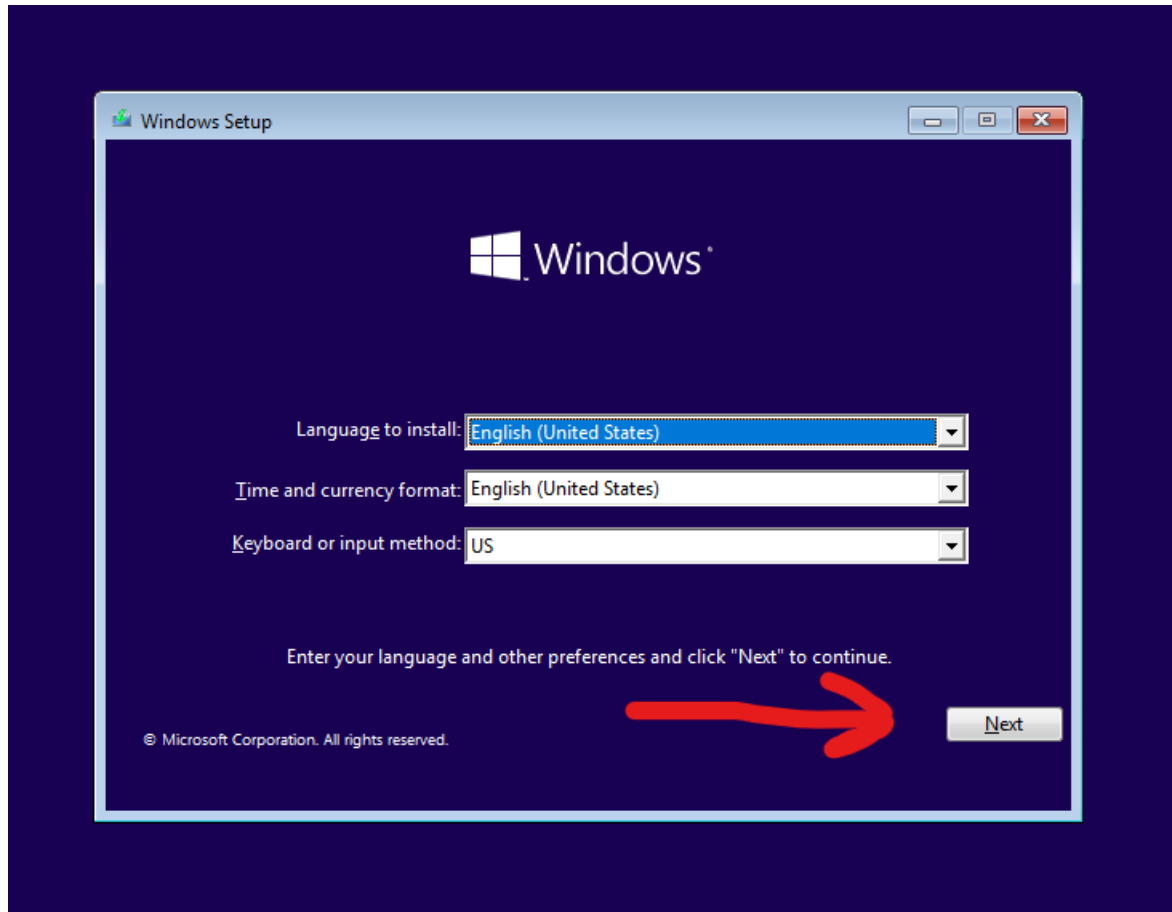


- Depending on your hypervisor, you should be greeted with a summary screen. Hit finish. Your new VM will populate on the left. Double click the new icon for your VM and let it start. Once it starts you will be greeted with a prompt to add a “DVD.” This is when we add our ISO.

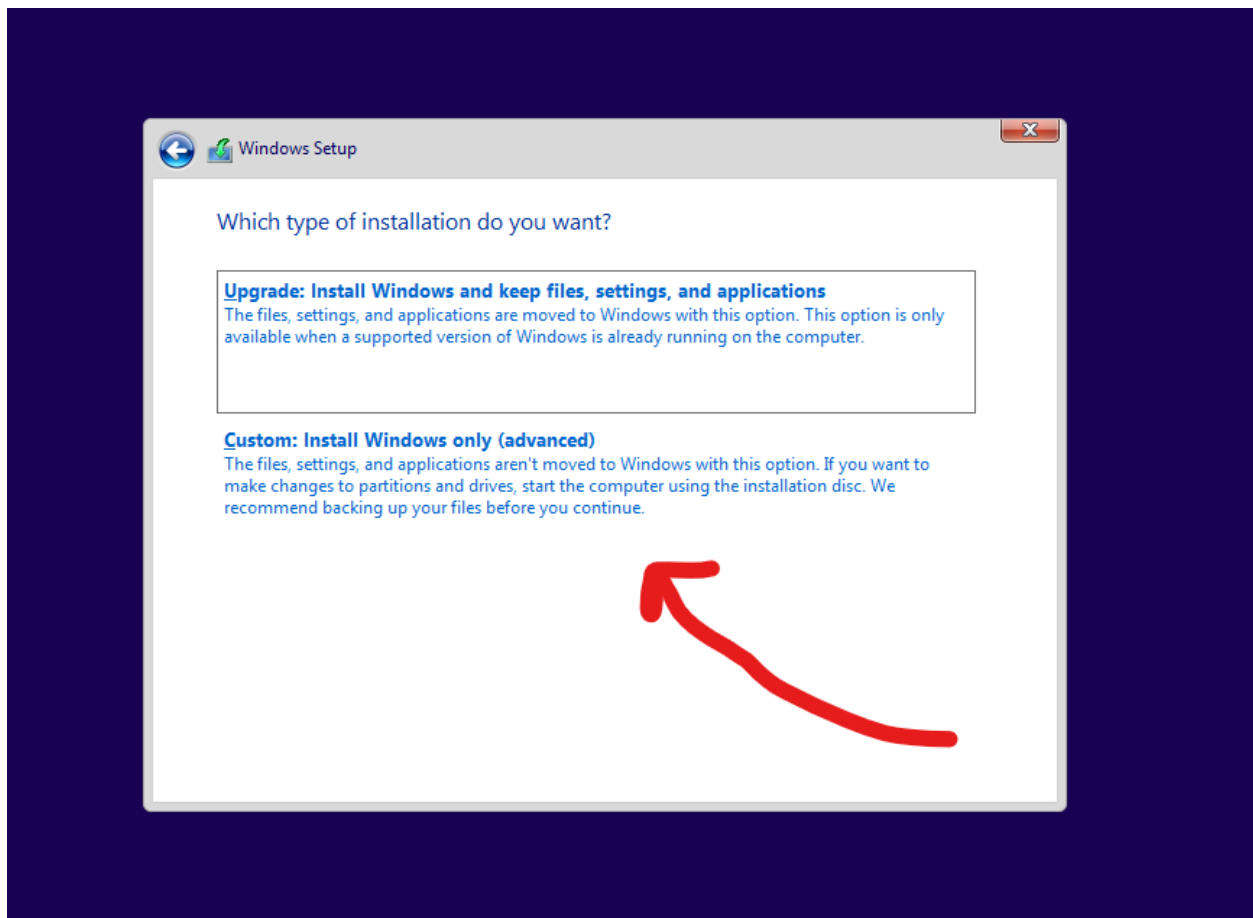


Click the down arrow and click “other,” your file browser will pop up. Navigate to the folder that you downloaded your Windows 10 ISO to and select it. Once it is selected, click “Mount and Retry Boot” in the bottom right.

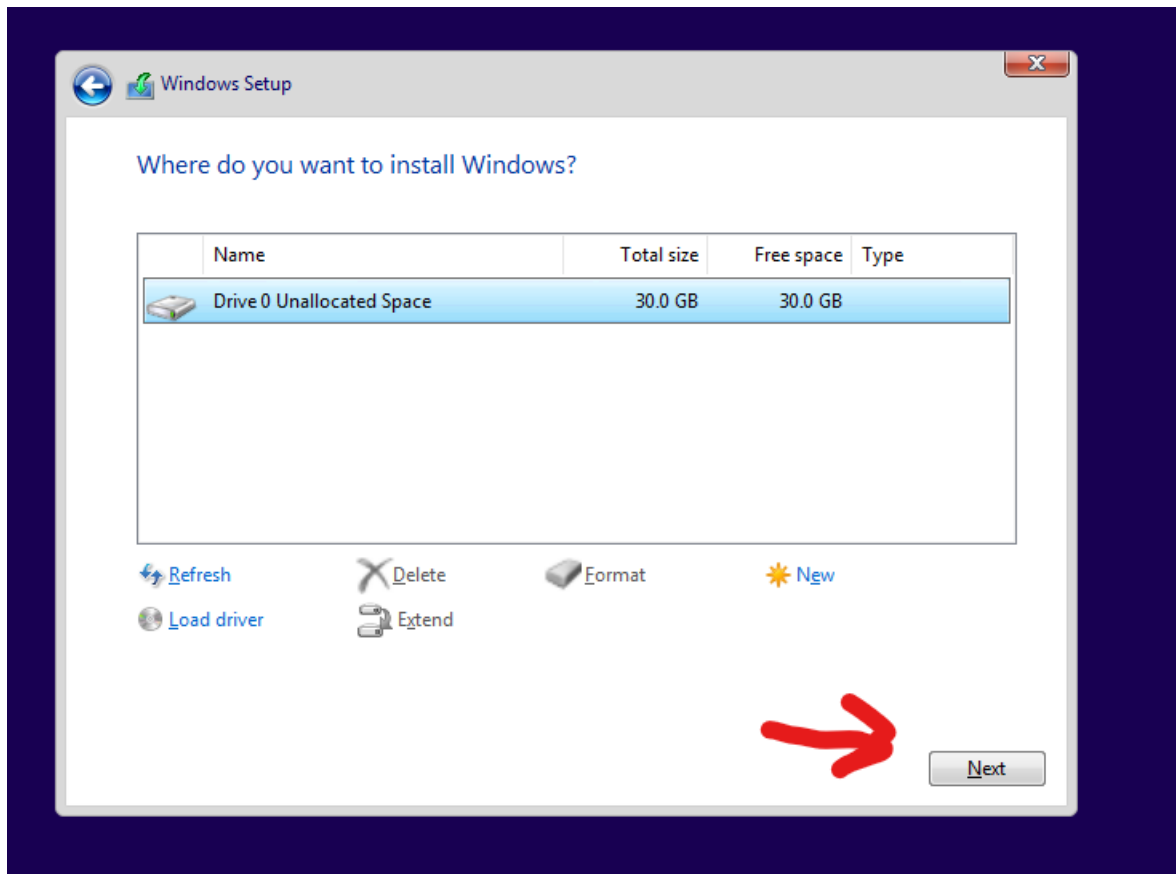
8. Once the VM reboots, Windows 10 Setup will load. Click next on the initial screen unless you want to change the language, time and currency format, or keyboard layout.



9. Accept the license terms and select “Custom: Install Windows only.”



10. Click next to choose the default virtual drive that we created earlier in the initial VM setup.



11. The OS will now install, it may take some time depending on the specs of your PC.

Windows Server 2022 Installation

While we wait for our Windows 10 installation to finish, we can start installing Windows Server 2022. All the steps above will be the same, the only difference is that you will choose Windows Server 2022 ISO instead of the Windows 10 one as well as the operating system. More details below.

Here are the steps to download the ISO:

1. Head over to this link: <https://www.microsoft.com/en-us/windows-server>

There will be a “Try Windows Server now” button. Once you click that, there will be another link to “Download free trial” of Windows Server 2022. After clicking that link, it will bring you to a page with the ISO download.

Get started for free

Please select your evaluation experience:

[Try Windows Server on Azure >](#)

[Create a Virtual Machine in Azure >](#)

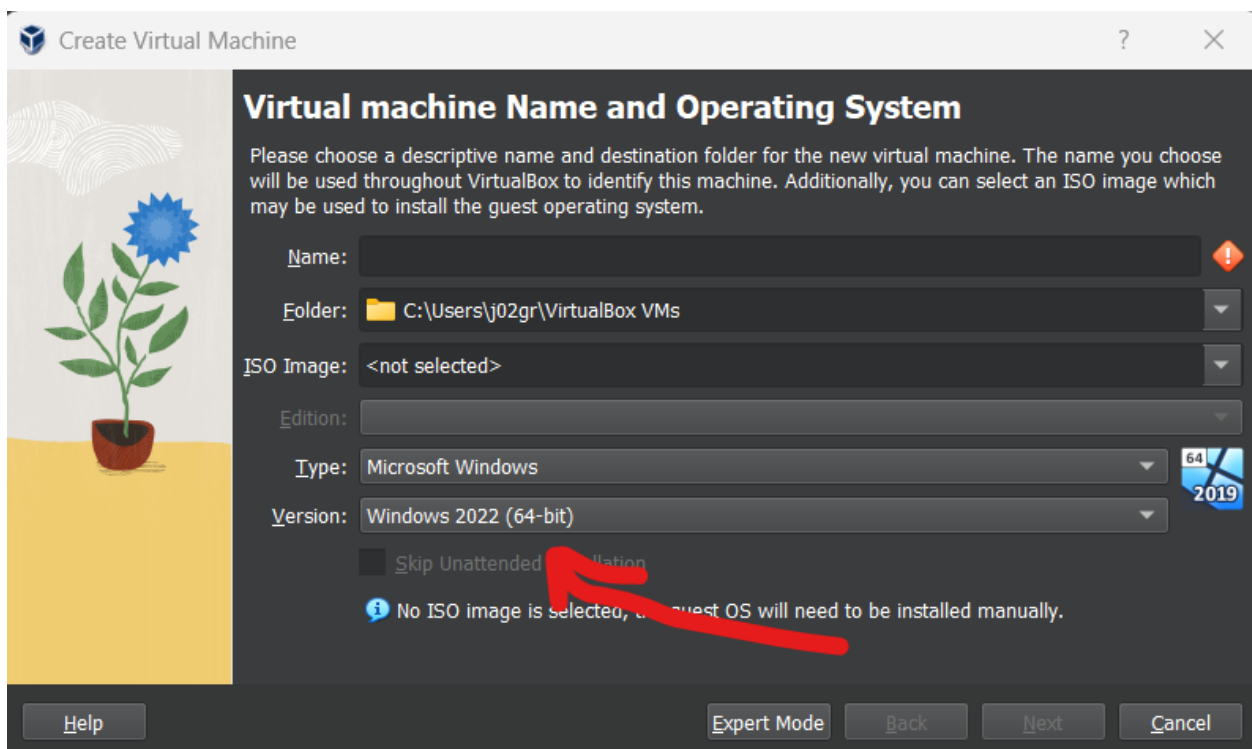
[Download the ISO >](#)

[Download the VHD >](#)

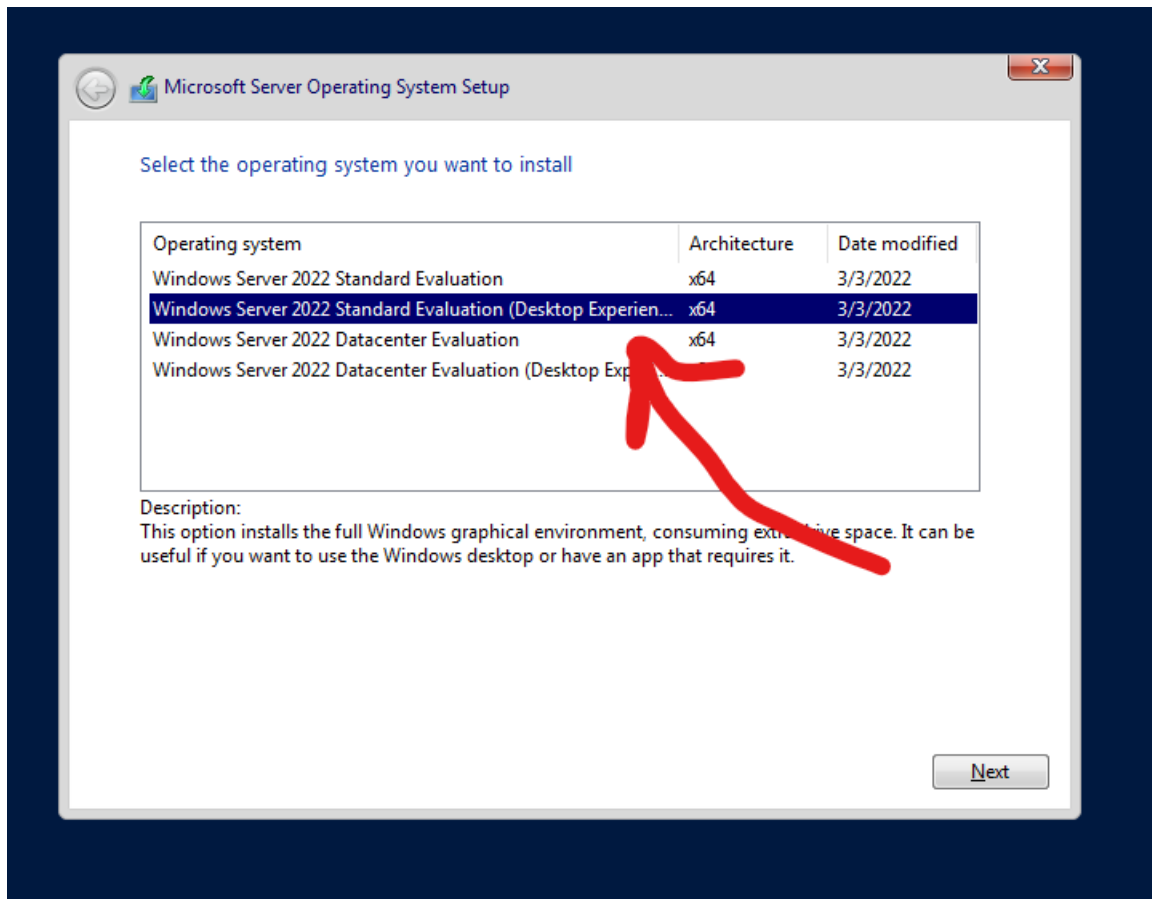


After filling out some details, you can download the ISO.

2. As I said before, all the steps will be the same as our first installation. Instead of choosing Windows 10 for the version when creating the VM, you will choose Windows 2022 (64-bit).



3. After you boot into the installation, you will get to a screen that asks you which operating system you want to install. Choose "Windows Server 2022 Standard Evaluation (Desktop Experience)."



4. Refer to our first installation of Windows 10 if you forgot any of the steps, and let the OS install.

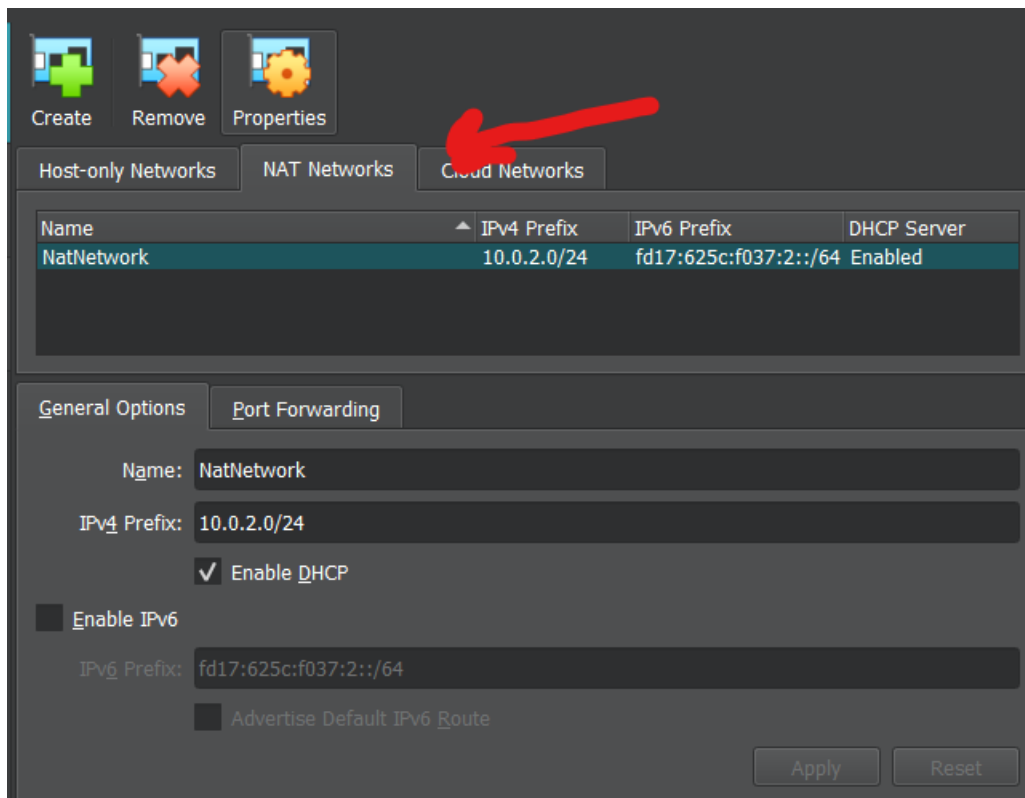
Miscellaneous Notes & Common Installation Issues

- We will configure the accounts later in this guide, if your installations are complete, continue below.
- If any of your installations fail at some point, you may not have configured the VM with enough disk space. Make sure you are allocating a minimum of 16 gigs.
- If you get an error about windows being unable to find the license when booting up the VM, that means you installed the ISO during the initial setup screen. Delete the VM, restart and do not install the ISO until you boot it up and it prompts you to select a DVD.

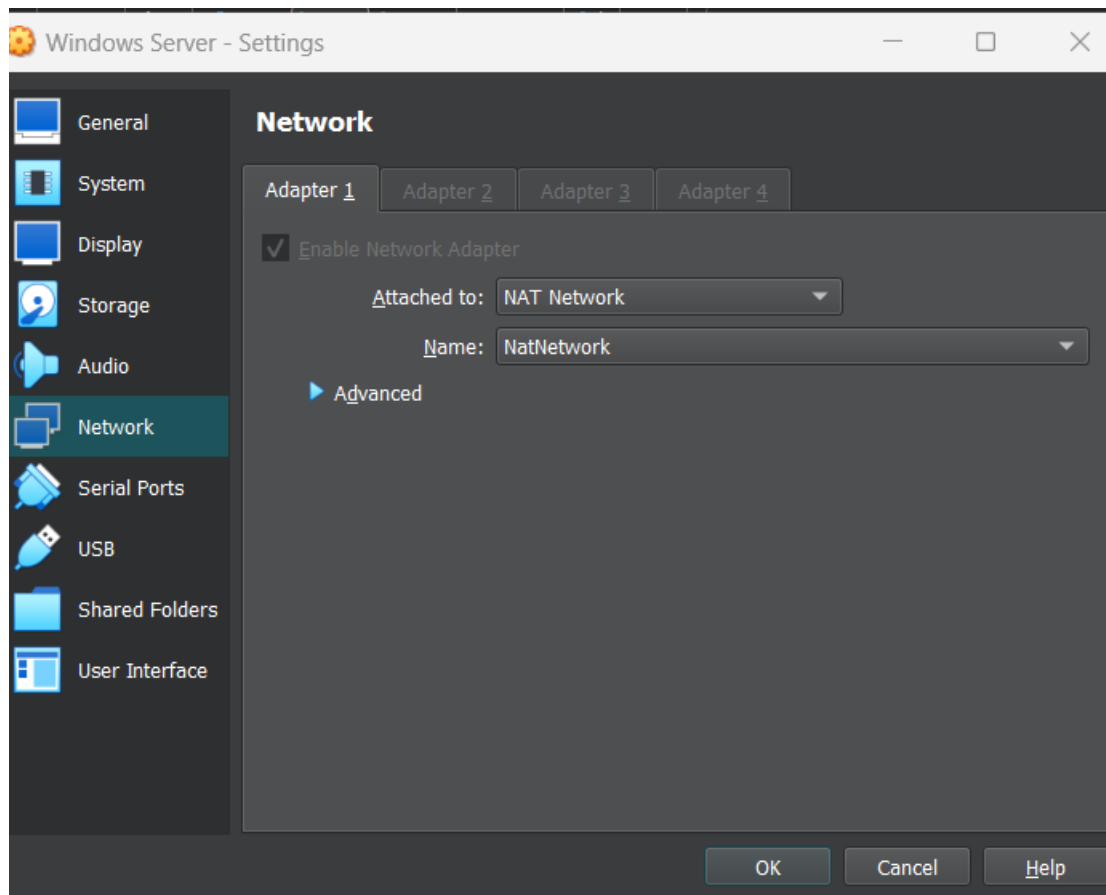
Configuring Our Virtual Network

We will now configure the isolated network that our VMs will be on. This is an important step as this is what allows us to use AD with our other VMs.

1. In the upper left-hand corner of VirtualBox, click File > Tools > Network Manager. Once you are in the Network Manager, head over to “NAT Networks.”



2. Copy the above settings. If you do not have a NatNetwork already, click create, and then fill in all the details from the image. Make sure DHCP is enabled.
3. Now that we have our NatNetwork created, we are going put our VMs on it. The steps will be the same for each VM.
4. Right click one of your VMs and click “settings.” Once you are in the settings panel, click on “Network” and change “Attached to:” to the NAT Network.



5. Repeat this step for your remaining VMs. Now you are all set! Your VMs are on their own isolated network.

Account Creation

We will now head back over to our VMs and create our accounts. We are going to start with our Windows 10 host(s).

1. If you have not already, go through the initial prompts. Choose your region, keyboard layout, and make your way to the Account screen. In the bottom left, you are going to click "Domain Join Instead."



Sign in with Microsoft

Work or school account

Sign in with a security key

Which account should I use?

Sign in with the username and password you use with Office 365 or other business services from Microsoft.



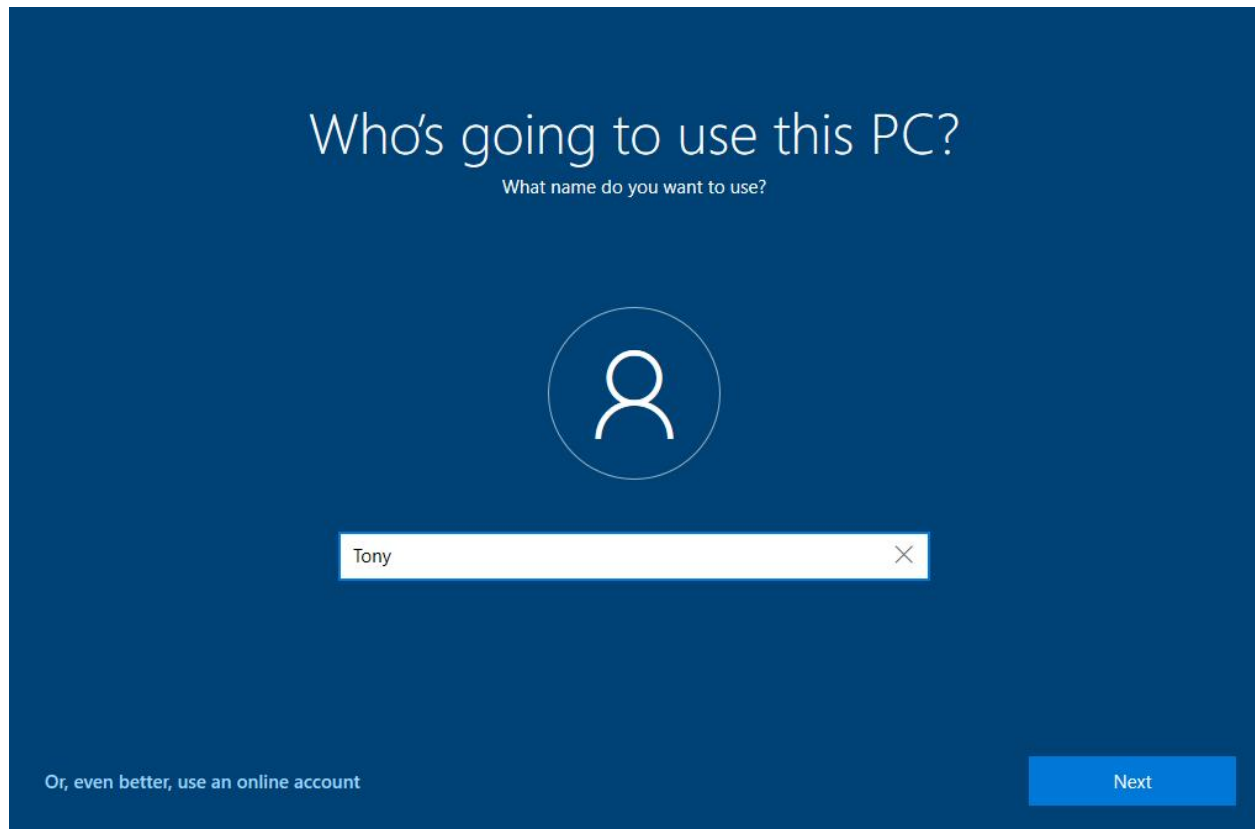
[Domain join instead](#)

[Privacy & cookies](#)

[Terms of use](#)

Next

2. Use whatever name suits your fancy and click next. (I chose tony for Tony Stark 😊)



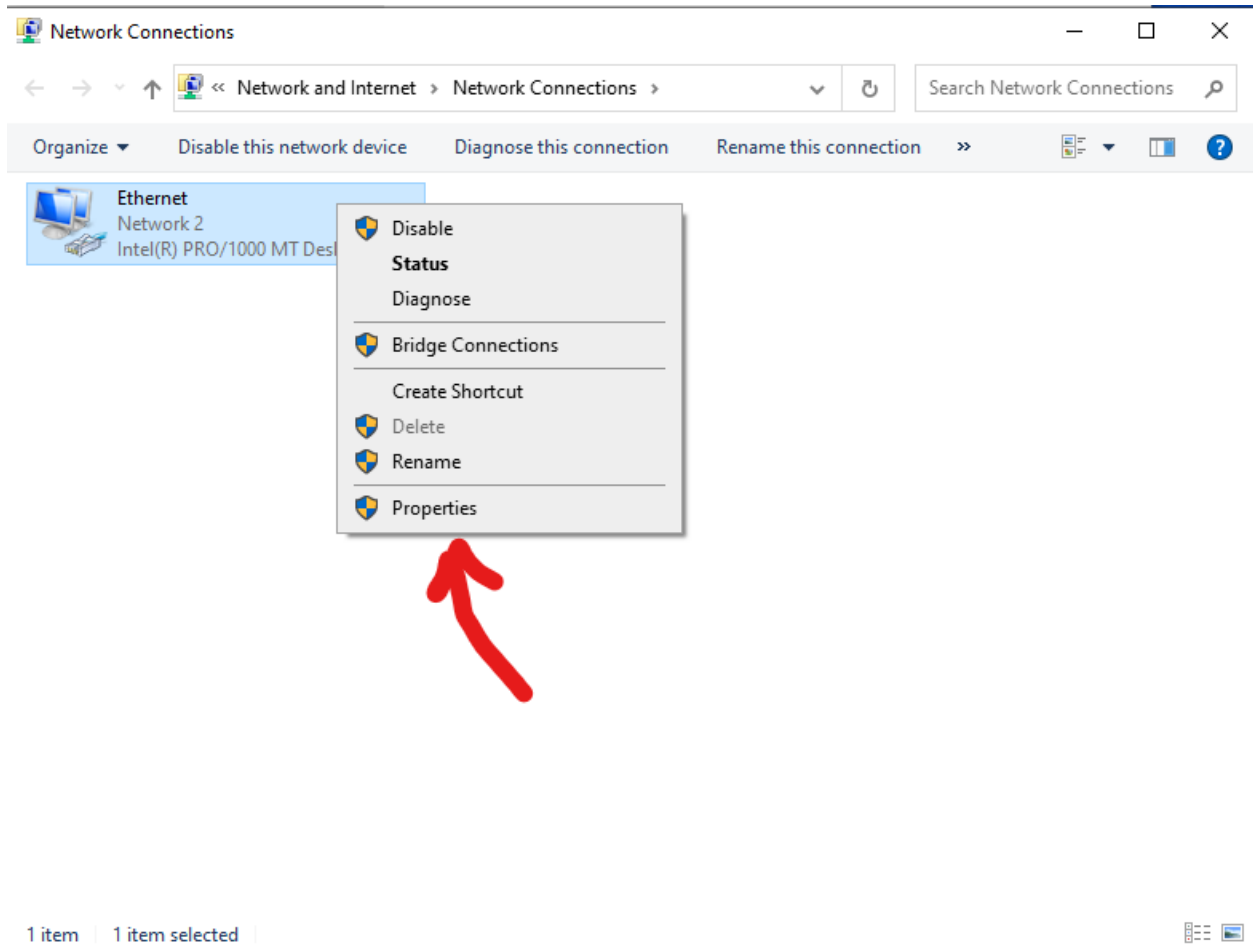
3. Enter in a password and fill out security questions. I disabled all the privacy settings, but it will not affect anything if you choose to leave them on.
4. Now repeat the process for your other windows hosts.

NOTE: For Windows Server, the only thing you need to do is enter a password.

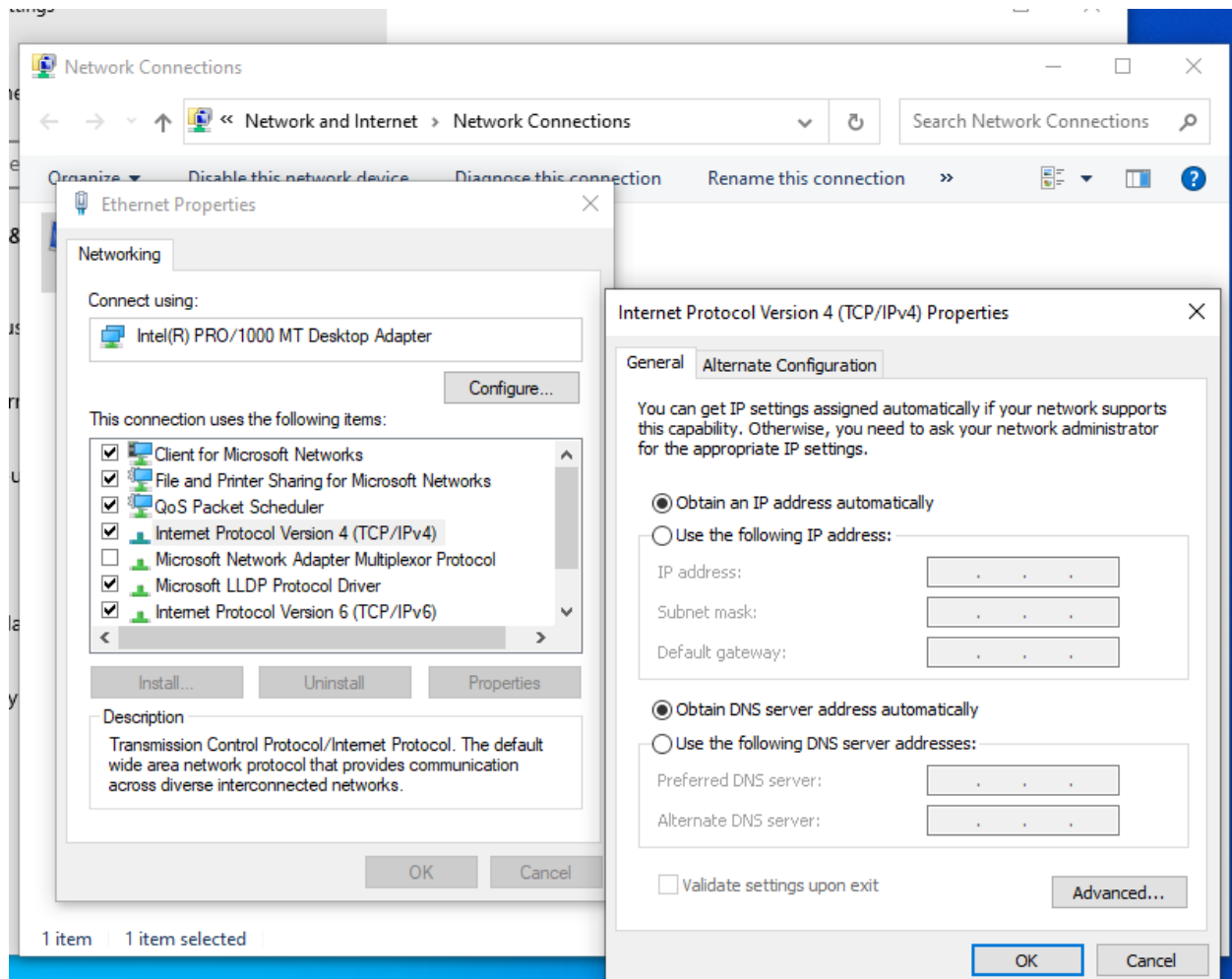
Configuring Our Hosts

After all that setup, we can finally get into the meat and potatoes. But there is one more thing we need to do in order for us to join our Windows hosts to the domain. We need to configure our Windows 10 hosts to use our Active Directory servers IP address as their DNS server.

1. On each of your Windows 10 hosts, navigate to settings > Network & Internet > change adapter options. Right click on your adapter and click "Properties."



2. On the properties page, click “Internet Protocol Version 4 (TCP/IPv4), and click “Properties.”



3. Now click “Use the following DNS server addresses.” And now we are going to head back over to our Windows Server 2022 VM.
4. Once you navigate back to you server VM, click the search bar and type “cmd.”
5. In the command prompt type “ipconfig”
6. Take note of the IPv4 address that is displayed on your server.

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ipconfig

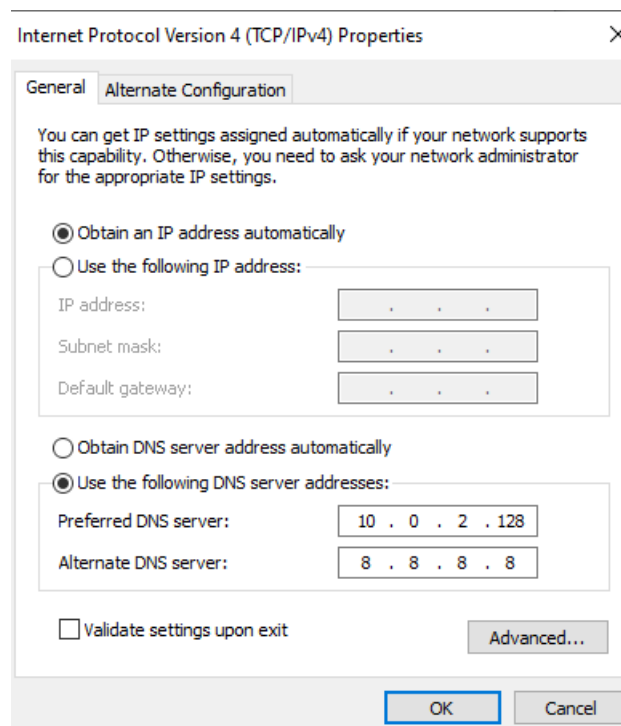
Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::3582:1532:190f:9aad%5
    IPv4 Address. . . . . : 10.0.2.128
    Subnet Mask . . . . . : 255.255.255
    Default Gateway . . . . . :

C:\Users\Administrator>
```

7. Now head back over to your windows 10 host and enter the IPv4 address from our server and enter it in the “Preferred DNS server” field.



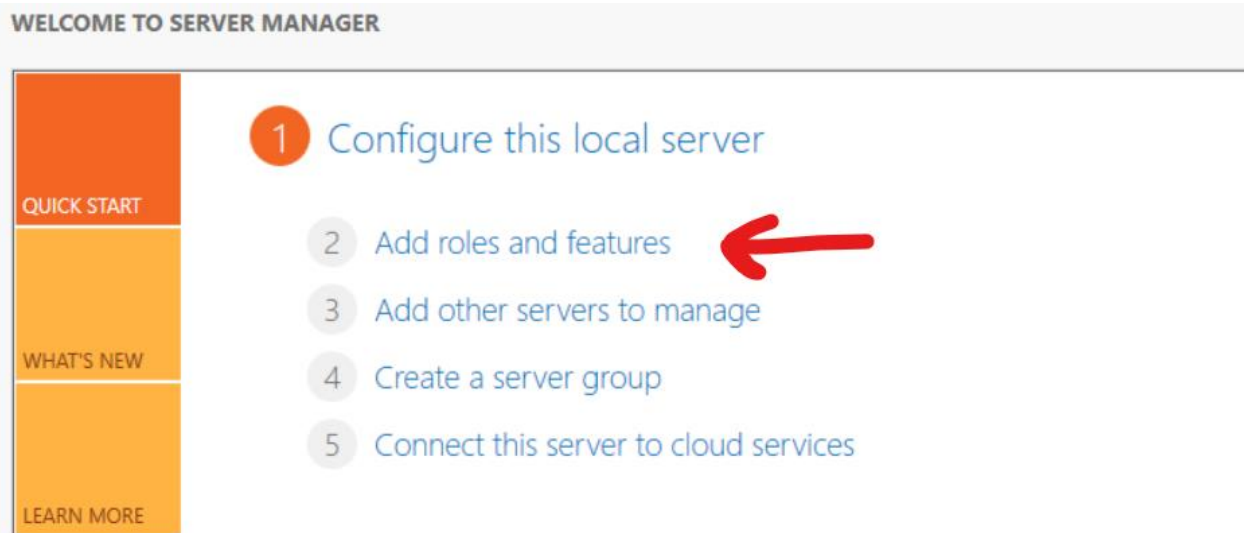
8. Repeat this process for however many other Windows 10 hosts you setup.

NOTE: You DO NOT need to do this for the server.

Active Directory Setup at Last

Okay, now I am serious, we can finally get into the meat and potatoes of this whole lab. Configuring Active Directory.

1. If you were observant, you should have noticed that an application called “Server Manager” booted when you first logged into your administrator account. If it did not, you can search for it in the search bar. From this page, we are going to install Active Directory. Click “Add roles and features.”



2. The wizard will pop up, from here we are going to click next until we get to the “Server roles” page. On this page, we are going to select “Active Directory Domain Services.”

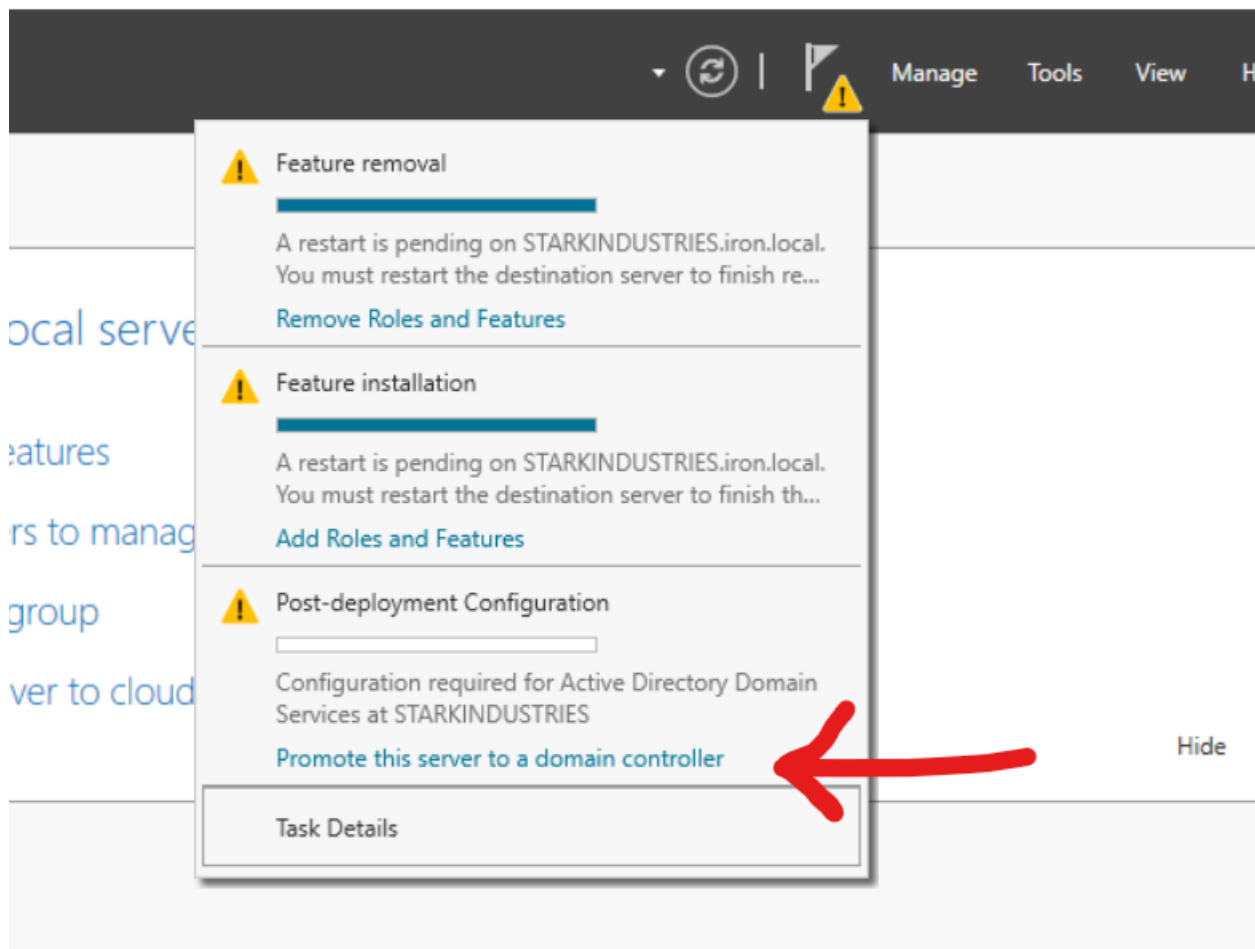
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results

Select one or more roles to install on the selected server.

Roles	Description
☐ Active Directory Certificate Services	Active Directory Domain Services (AD DS) stores information about objects on the network and makes this information available to users and network administrators. AD DS uses domain controllers to give network users access to permitted resources anywhere on the network through a single logon process.
☐ Active Directory Domain Services	
☐ Active Directory Federation Services	
☐ Active Directory Lightweight Directory Services	
☐ Active Directory Rights Management Services	
☐ Device Health Attestation	
☒ DHCP Server (Installed)	
☒ DNS Server (Installed)	
☐ Fax Server	
☒ File and Storage Services (2 of 12 installed)	
☐ Host Guardian Service	
☐ Hyper-V	
☐ Network Policy and Access Services	
☐ Print and Document Services	
☐ Remote Access	
☐ Remote Desktop Services	
☐ Volume Activation Services	
☐ Web Server (IIS)	

< Previous
Next >
Install
Cancel

- Once you check the box, a window will pop up. From here, click "Add Features."
- Click next and leave the features default.
- You will get to the confirmation page, click install in the bottom right and let feature installation finish.
- Active Directory is now installed, in the top right corner, click the flag with the yellow exclamation point triangle, and click "promote this server to a domain controller."



7. A new window will pop up. We need to create a new forest. Click the “Add a new forest” button and specify a root domain name. I will call mine “iron.local” but you can name yours whatever you like.

MAKE SURE YOU ADD THE .local AFTER WHATEVER YOU NAMED IT

Deployment Configuration

TARGET SERVER
STARKINDUSTRIES.iron.local

Deployment Configuration

Domain Controller Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select the deployment operation

- ☐ Add a domain controller to an existing domain
- ☐ Add a new domain to an existing forest
- ☒ Add a new forest

Specify the domain information for this operation

Root domain name:

[More about deployment configurations](#)

< Previous

Next >

Install

Cancel

8. Click next and on the Domain Controller Options page, specify a DSRM password. Leave everything else default.

Domain Controller Options

TARGET SERVER
STARKINDUSTRIES.iron.local

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select functional level of the new forest and root domain

Forest functional level:

Windows Server 2016

Domain functional level:

Windows Server 2016

Specify domain controller capabilities

☒ Domain Name System (DNS) server☒ Global Catalog (GC)☐ Read only domain controller (RODC)

Type the Directory Services Restore Mode (DSRM) password

Password:

••••••••••••••••

Confirm password:

••••••••••••••••

[More about domain controller options](#)

< Previous

Next >

Install

Cancel

- Click next, not selecting the “Create DNS Delegation” button until you reach the “Additional Options” page. On this page, the NETBIOS domain name will be generated. It should be whatever you put for the root domain name without the .local at the end.

Active Directory Domain Services Configuration Wizard

Additional Options

TARGET SERVER
STARKINDUSTRIES.iron.local

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Verify the NetBIOS name assigned to the domain and change it if necessary

The NetBIOS domain name:

[More about additional options](#)

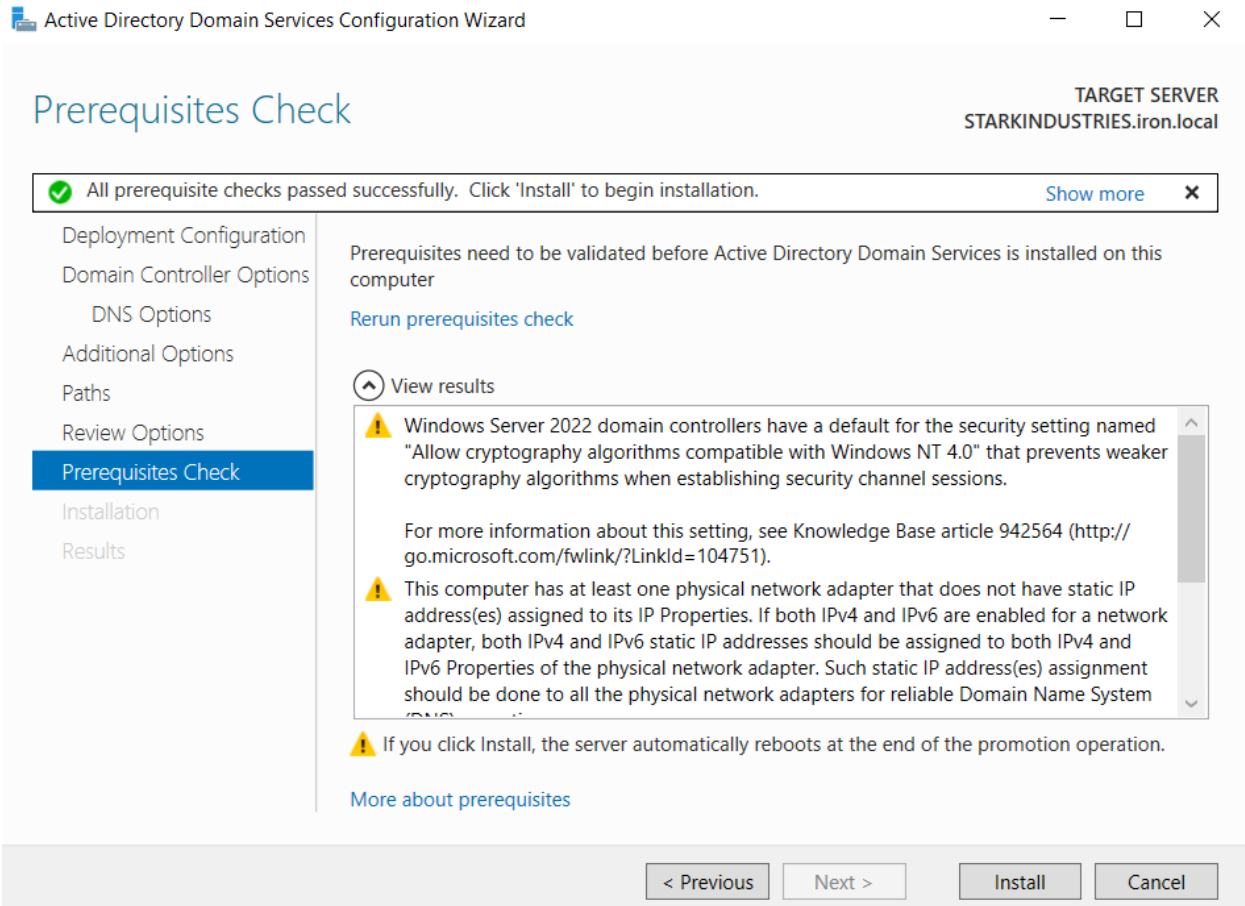
< Previous

Next >

Install

Cancel

- Click next until you reach the prerequisites check page, leaving everything default. On the prerequisite check page, it will check the system to make sure it can successfully install AD.

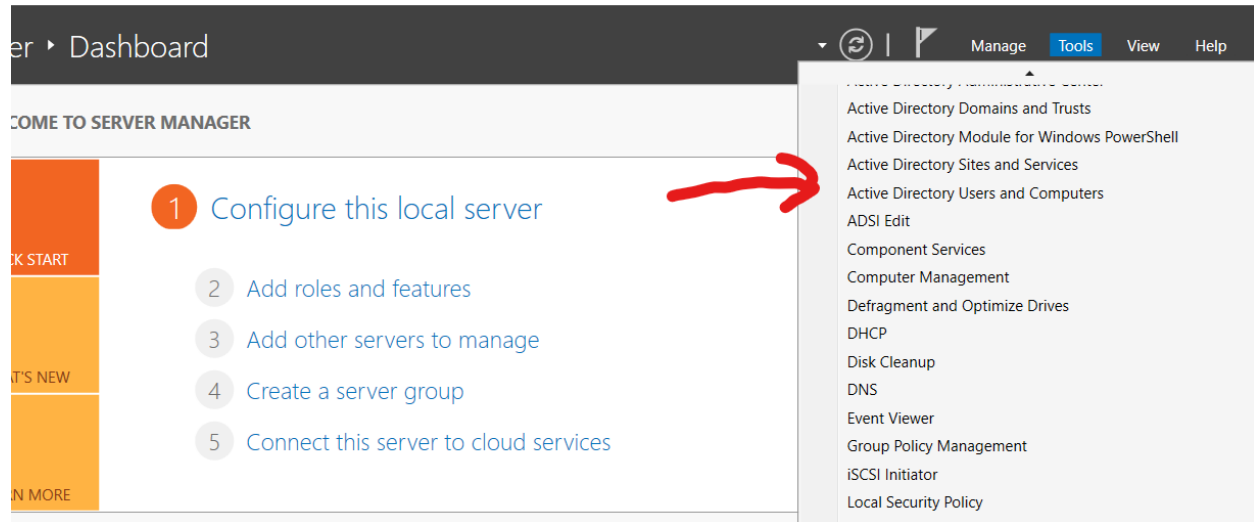


11. Once the check is completed, click install. After the installation is completed, you will need to reboot the server. It may take a while for the settings to apply.

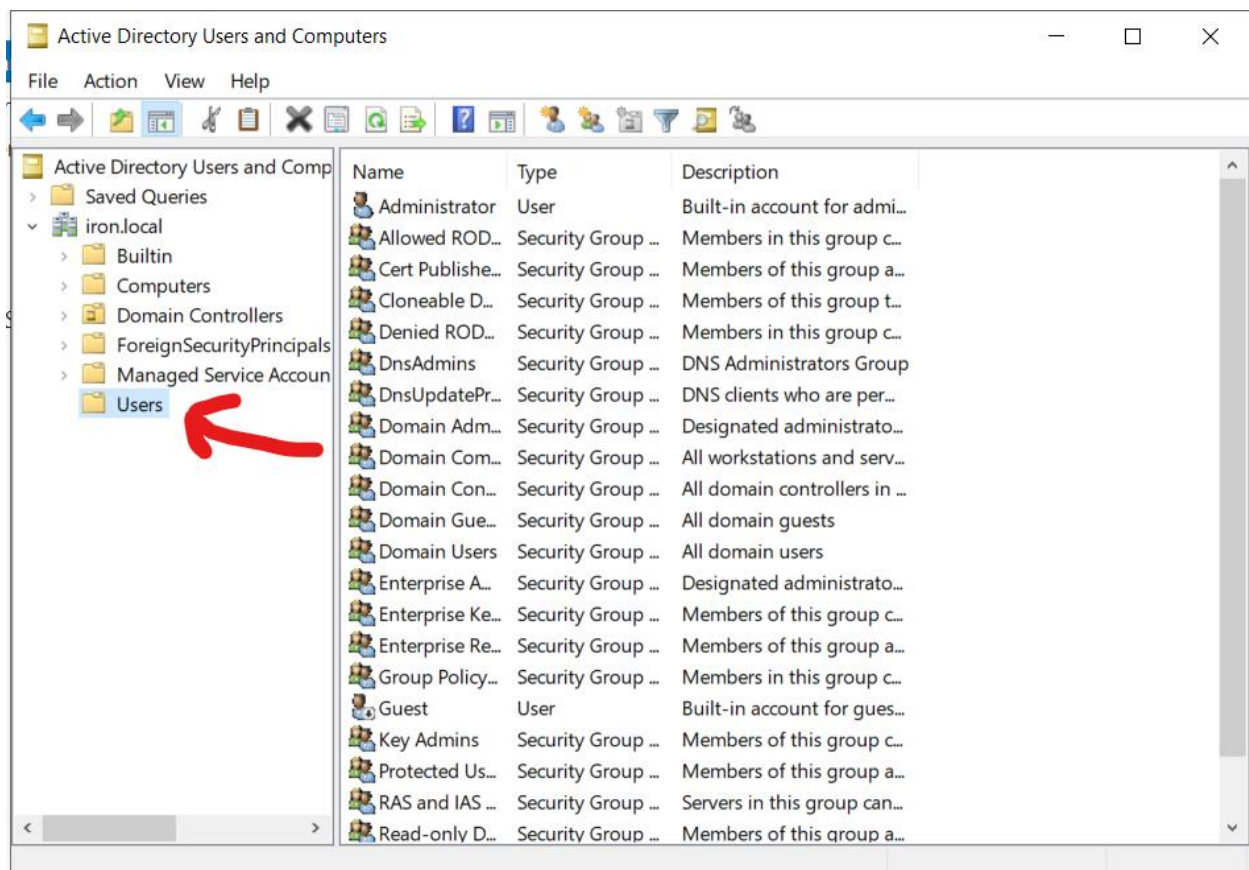
Creating Users in Active Directory

Now that we have finished installing Active Directory, we can now create some users.

1. In the top right of Server Manager, Navigate to Tools > Active Directory Users and Computers.



2. Click the arrow next to your domain and navigate to the users section.



3. In the users panel, right click and select New > User. A user object creation box will pop up.

New Object - User ×

 Create in: iron.local/Users

First name: Initials:

Last name:

Full name:

User logon name: @iron.local ▼

User logon name (pre-Windows 2000):

< Back Next > Cancel

4. Fill in whatever first and last name you would like. I will be doing “Tony Stark.” For the username, first initial and last name of your user and click next.

New Object - User



Create in: iron.local/Users

First name: Initials:

Last name:

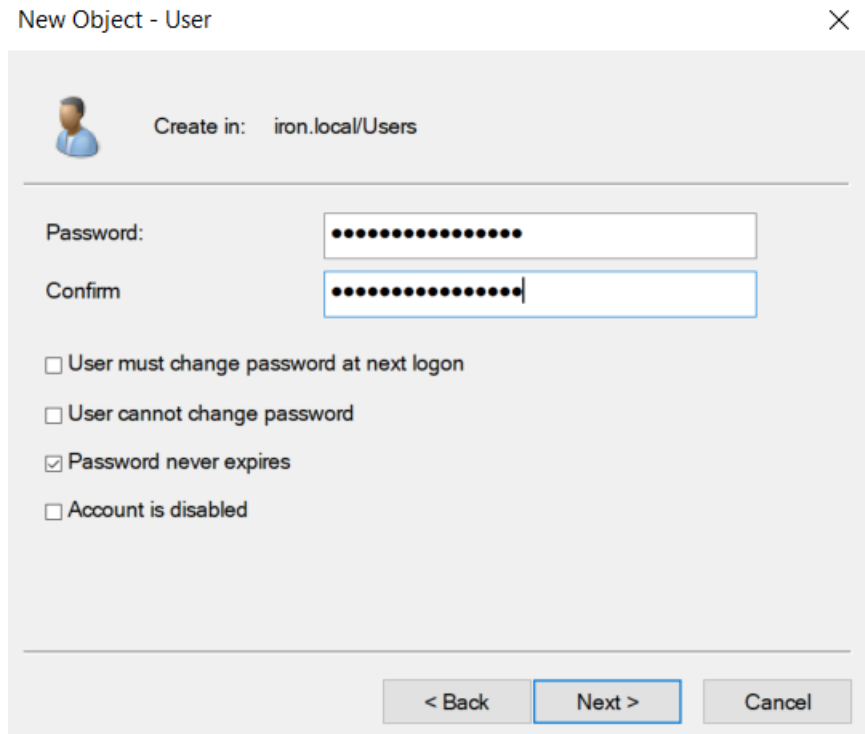
Full name:

User logon name:

User logon name (pre-Windows 2000):

5. Specify a password for the user and uncheck the “User must change password at next logon” option. Select the “password never expires” option.

NOTE: these settings are just to make our lives easier. We are in an isolated lab environment, in a normal enterprise environment, you would not set the password to never expire, and you would keep the option selected for letting the user change the password on next login.



New Object - User

Create in: iron.local/Users

Password: [password field]

Confirm [password field]

☐ User must change password at next logon

☐ User cannot change password

☒ Password never expires

☐ Account is disabled

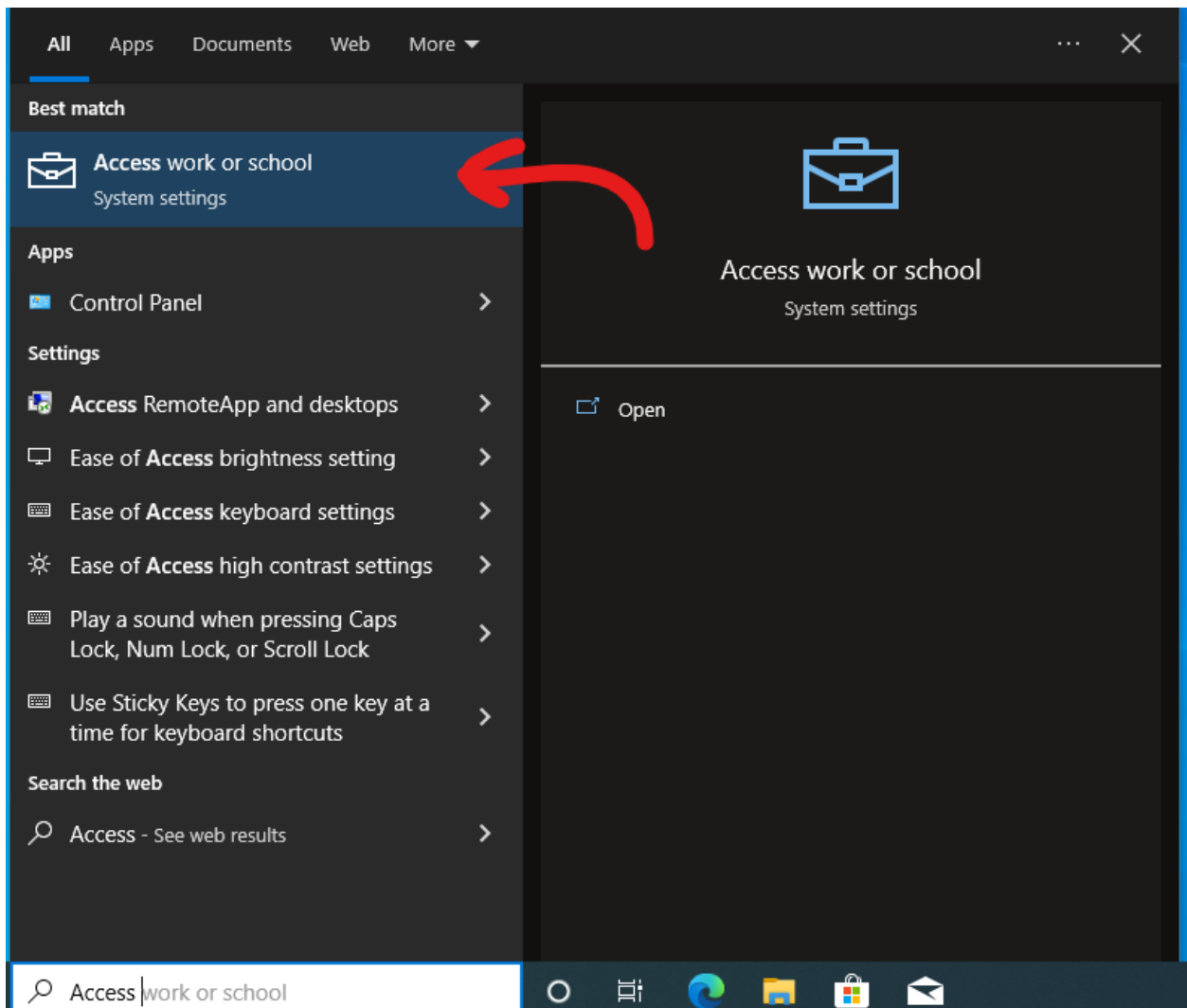
< Back Next > Cancel

6. Click next and confirm that the details are correct on the summary screen.
7. Congrats! You created your first user in AD. Repeat this process for however many host VMs you decided to use for your lab.

Joining Our Hosts to the Domain

At this point in the lab, you should have created the number of users respective to the amount of VMs you have. We will now join each VM to the domain.

1. Navigate to one of your Windows 10 hosts.
2. In the search bar type "access work or school"



3. Once the window pops up, click “connect now.” Instead of entering a microsoft account, you are going to select “join this domain to a local Active Directory domain” in the bottom right.

NOTE: If you configured your host’s preferred DNS server with your servers IP address, you should get a prompt to login. If not, refer to the “Configuring Our Hosts” section of the lab.

Access work or school

Get access to resources like email, apps, and the network. Connecting means your work or school might control some things on this device, such as which settings you can change. For specific info about this, ask them.

 Connect

Microsoft account

Set up a work or school account

You'll get access to resources like email, apps, and the network. Connecting means your work or school might control some things on this device, such as which settings you can change. For specific info about this, ask them.

Alternate actions:

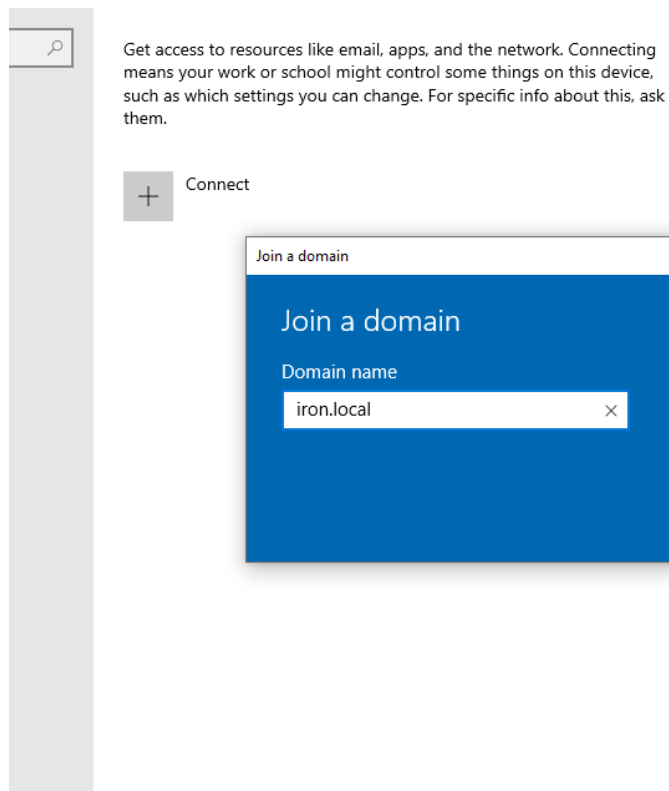
[Join this device to a local Active Directory domain](#)

Next

Re
Ac
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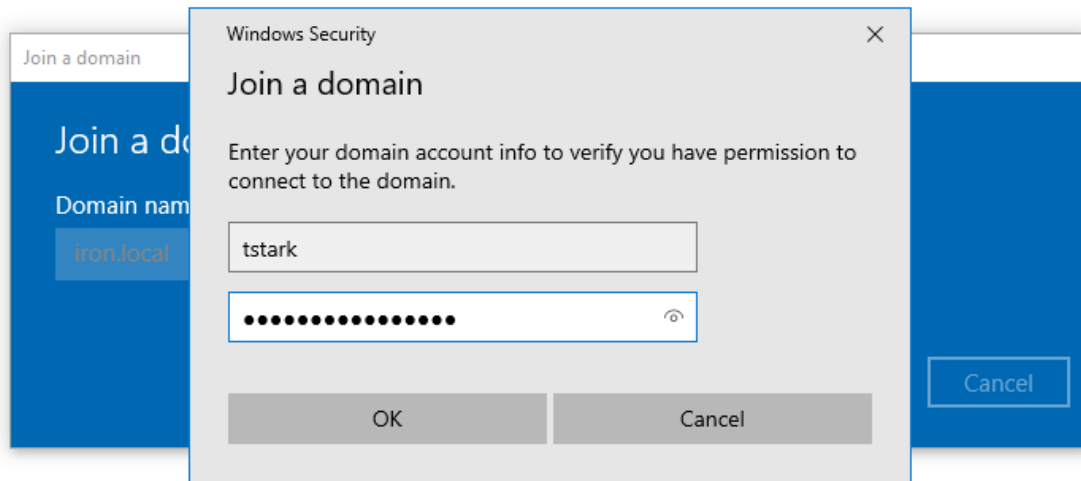
4. After selecting the local AD option, you will be prompted to join a domain. From here, you will type in the name that you specified for your root domain (i.e. example.local)



5. If you typed in your domain correctly, you will be asked to login. Choose one of the accounts that you created within the Active Directory users section from earlier.

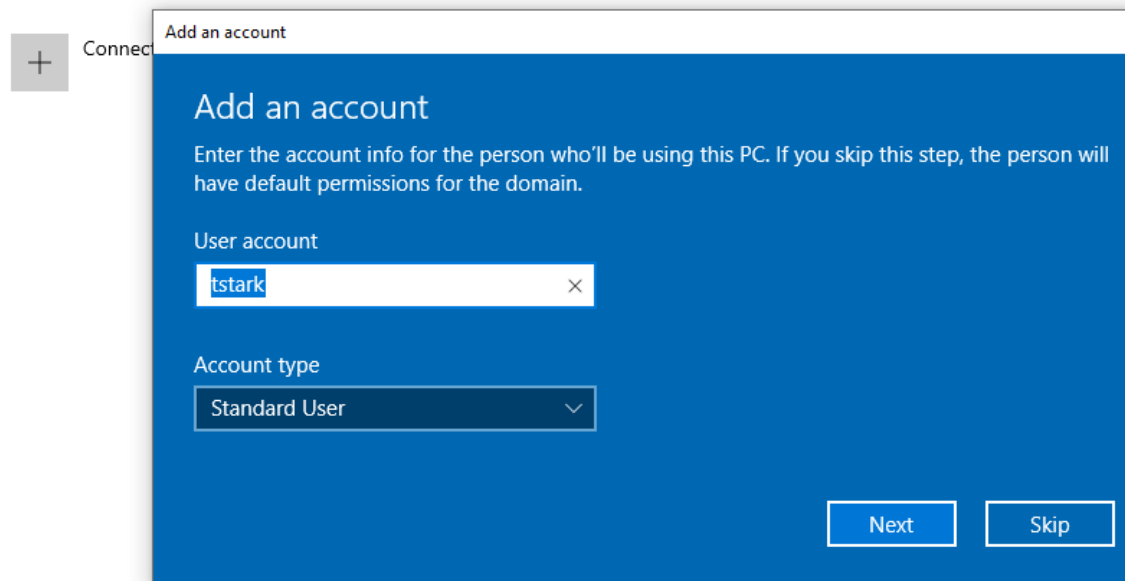
which settings you can change. For specific info about this, ask

Connect



6. If you specified a correct account, a box will pop up asking you to add an account. Leave the user account field default and make the account a "Standard User."

Get access to resources like email, apps, and the network. Connecting means your work or school might control some things on this device, such as which settings you can change. For specific info about this, ask them.



+

Connect

Add an account

Add an account

Enter the account info for the person who'll be using this PC. If you skip this step, the person will have default permissions for the domain.

User account

tstark

Account type

Standard User

Next

Skip

7. Go ahead and restart the pc when prompted.
8. Repeat this process for each Windows 10 host you have.

Conclusion

Congratulations! You did it! You properly installed and configured Active Directory. The whole point of this lab was to get your feet wet with AD. Hopefully now you have a better understanding of how it works. Feel free to mess around a bit with it, explore, change some settings, configure some things. After all, that is the whole point of a home lab!

I have lots of plans for future labs, some more stuff with Active Directory, configuring a SIEM, and doing some vulnerability scanning and exploitation, and some pfSense firewall stuff. I hope you found this useful, if you have any questions, feel free to reach out to me on LinkedIn.